Semantic-Guided Imitation Learning

***Abstract*—**

***Index Terms*—Imitation learning, semantic representations, visuomotor control.**

I. Introduction

Reliable execution of sequential manipulation tasks is essential for practical robotic applications but remains challenging [1]. Sequential manipulation tasks require a robot to achieve an overall goal through a sequence of interrelated operations [2]. Such tasks arise in practical applications ranging from household cleaning, where a tablecloth is lifted to clear debris beneath it and then restored, to surgery, where tissue is retracted to expose the target region before resection [3]. A key challenge of these tasks is that preceding interactions change the conditions under which subsequent operations can be performed. For example, compared with simple pick-and-place tasks involving visible and accessible targets, manipulation under occlusion requires the robot to assess whether prior actions have made the target visible and accessible or further manipulation is needed [4], [5]. Consequently, reliable execution requires a policy that adapts subsequent actions to interaction outcomes observed through visual feedback.

Visuomotor imitation learning offers a practical approach to sequential manipulation by reducing the need for explicit interaction modeling, but reliable execution remains difficult with limited demonstrations. It learns a policy that maps visual and proprioceptive observations to actions using expert demonstrations. For example, Action Chunking with Transformers (ACT) [6] has been used in Mobile ALOHA to learn coordinated mobile manipulation, including opening a cabinet and storing a cooking pot [7]. Similarly, Diffusion Policy [8] has been adopted in the Universal Manipulation Interface (UMI) framework to learn bimanual cloth folding and multi-step dish washing [9]. However, execution errors can lead a policy into states poorly covered by limited demonstrations, where its predicted actions may no longer suit the actual task conditions, compromising subsequent operations and task completion [10].

Prior work has focused on improving the robustness of visuomotor policies to execution deviations by expanding data coverage, improving visual representations, and adapting actions during execution. Ross et al. [10] expanded the training dataset with expert action labels for states visited by the learned policy, while Laskey et al. [11] injected noise into expert actions during demonstration collection to obtain examples of recovery behavior. Although data coverage expansion improves coverage of states arising from execution deviations, it requires additional expert supervision or modifications to demonstration collection, limiting its applicability when only a fixed demonstration set is available. Zhu et al. introduced object-centric visual representations for robust visuomotor imitation [12] and incorporated 3D object representations to improve generalization across backgrounds, viewpoints, and object instances [13]. Although these methods improve generalization to visual variations, robust object representations alone do not ensure that subsequent actions remain appropriate when execution deviates from demonstrations.

In this work, we develop a theory-guided visuomotor imitation learning framework that uses task-relevant representation learning to support sequential manipulation from limited demonstrations. Our analysis derives conditions linking imitation error and representation quality to task-failure bounds. Guided by this analysis, we supervise policy query tokens with task descriptors derived from semantic masks, encouraging the policy to retain information relevant to changing manipulation conditions. The policy is trained from the fixed demonstration set, without additional robot interaction for policy training. Simulation and real-world experiments demonstrate the effectiveness of the proposed framework in sequential manipulation tasks.

II. Related Work

III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. Starting from the relationship between imitation error and task success, we examine how to preserve information needed to distinguish the task consequences of different actions.

We first establish conditions under which action error bounds task-failure probability and explain why improved coverage in representation space alone does not guarantee better task performance. We then examine how semantic guidance can preserve task-relevant information and, under consistent expert selection, control imitation-error transfer through expert-action reconstruction.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available execution history. Training is based on this fixed demonstration dataset, with demonstrated actions providing the primary imitation target, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) System and Observations: Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t. The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images

$I_t^{1:V} = (I_t^1, \ldots, I_t^V)$ and proprioceptive information q_t, where $V \geq 1$ is the number of cameras and I_t^v is the image from the vth view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \tag{1}$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) Offline Demonstrations and Policy: The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The ith demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{\mathrm{E},i}, o_1^i, \ldots, u_{T_i-1}^{\mathrm{E},i}, o_{T_i}^i). \tag{2}$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the ith trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π is represented by the history $\mathcal{H}_t$. This history includes observations $o_{0:t}$ obtained up to time t, previously executed commands $u_{0:t-1}$, and other runtime records before the current decision. The policy defines a distribution over admissible control commands, with $u_t \sim \pi(\cdot \mid \mathcal{H}_t)$. The command history is empty at $t = 0$.

3) Learning and Task Objectives: For the following analysis, we consider deterministic policies acting on the available history, with π_{E} denoting the expert policy. Let ν denote the population distribution underlying the demonstrations, where $(t, h) \sim \nu$ specifies a control step and its available history. We define the population imitation risk as the expected squared Euclidean deviation from the expert's action:

$$R_\nu(\pi) := \mathbb{E}_{(t,h)\sim\nu} \|\pi(h) - \pi_{\mathrm{E}}(h)\|^2. \tag{3}$$

Training seeks to reduce this risk through an empirical imitation loss on the finite demonstrations; the specific training losses are defined in Section IV.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \to \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π:

$$\Psi = \Phi(s_{0:T}), \qquad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \tag{4}$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π. Learning aims to reduce $R_\nu(\pi)$, whereas deployment aims to increase $J(\pi)$. A lower imitation risk does not by itself guarantee a higher task-completion probability; the following analysis examines conditions relating these objectives.

4) Visual Assessability: Under a fixed camera configuration and evaluation horizon T, the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \to \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \tag{5}$$

Assumption 1 specifies visual identifiability of completion within the evaluation domain, without requiring full-state observability. The subsequent analysis considers policy executions and counterfactual action comparisons for which $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$ almost surely.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and show that transferring imitation risk from demonstrations to deployment depends jointly on representation-level coverage and conditional action-error discrepancy.

1) Task-Performance Difference Between Policies: We compare a policy π with a reference policy π_0 through their task-failure probabilities and express the difference as a sum of expected single-step action consequences. Each consequence is defined by comparing alternative actions at the same history while following π_0 afterward [14], [15]. This decomposition provides the basis for relating imitation error to task performance.

We consider deterministic policies acting on the available history. Given a history $\mathcal{H}_t = h$ and a candidate action $a \in \mathcal{A}$, define the action-value function as the probability of task failure ($\Psi = 0$):

$$Q_t^{\pi_0}(h, a) := \mathbb{P}^{a,\pi_0}(\Psi = 0 \mid \mathcal{H}_t = h)\,. \tag{6}$$

Here, $\mathbb{P}^{a,\pi_0}$ denotes the probability law obtained by taking action a at time t and following π_0 afterward.

When the current action is also selected by π_0, this reduces to the value function of π_0, $V_t^{\pi_0}(h) := Q_t^{\pi_0}\big(h, \pi_0(h)\big)$. The effect of choosing a instead of $\pi_0(h)$ is therefore measured by

$$A_t^{\pi_0}(h, a) := Q_t^{\pi_0}(h, a) - V_t^{\pi_0}(h). \tag{7}$$

Thus, $A_t^{\pi_0}(h, a) > 0$ indicates an increase in failure probability, whereas $A_t^{\pi_0}(h, a) < 0$ indicates a decrease.

To relate these action-level consequences to differences in task performance between policies, we require the following common comparison conditions.

Assumption 2 (Common task and environment): The compared policies share the same initial distribution and task-completion criterion. Given a history h and an action a, the next available history follows a conditional distribution that does not otherwise depend on the policy. The reference policy π_0 is defined on all histories involved in the comparison.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π. To evaluate π relative to π_0, we set $a = \pi(h)$ and aggregate the resulting action consequences along trajectories generated by π.

Applying the finite-horizon performance-difference identity [14]–[16] to these one-step action effects gives

$$F(\pi) - F(\pi_0) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) \right]. \tag{8}$$

Here, $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π. Equation (8) evaluates action effects at the situations actually visited by the evaluated policy π. Its proof is given in Appendix A-A. This relation allows task-performance differences to be bounded through the consequences of action deviations.

2) Task-Failure Bounds from Imitation Risk: Taking the expert policy as the reference, we show that the imitation risk $R_\nu(\pi)$ bounds task-failure probability $F(\pi)$ when demonstrations sufficiently cover deployment histories and action deviations have bounded task consequences.

Imitation risk is evaluated under the demonstration distribution ν, but deployment histories may follow a different distribution because actions depend on π. Let μ_π^t denote the distribution of $(t, \mathcal{H}_t)$ at control step t under π. The distribution $\bar{\mu}_\pi = T^{-1} \sum_{t=0}^{T-1} \mu_\pi^t$ corresponds to choosing a control step uniformly and observing its history.

Assumption 3 (Demonstration coverage):

For the evaluated policy π, there exists a uniform constant $1 \le C < \infty$ such that

$$\bar{\mu}_\pi(\mathcal{B}) \le C\, \nu(\mathcal{B})$$

for every measurable set $\mathcal{B}$ of pairs (t, h).

This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Assumption 4 (Action sensitivity): Let A_t^{E} denote the action effect with the expert policy as the reference. For a fixed action scale, there exists a uniform constant $0 \le L < \infty$ such that

$$|A_t^{\mathrm{E}}(h, a)| \le L \|a - \pi_{\mathrm{E}}(h)\|$$

for all relevant (t, h, a).

This assumption allows action deviations to be translated into bounds on changes in task-failure probability.

Following the connection between deployment imitation error and task cost [10], Assumptions 3 and 4, together with the performance-difference identity, yield the following bound (Appendix A-B):

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C R_\nu(\pi)}. \tag{9}$$

Here, C accounts for the shift from demonstration to deployment distributions, while L measures the sensitivity of task outcomes to action deviations. With other quantities fixed, a smaller $R_\nu(\pi)$ yields a tighter upper bound on task-failure probability.

This bound identifies coverage and action sensitivity as key conditions for translating imitation accuracy into task-performance guarantees. We next examine the limitations of complete-history coverage and the role of representations in risk transfer.

3) Imitation-Risk Transfer Through Representations: To address the limitations of complete-history coverage, we analyze imitation-risk transfer in representation space. We derive a bound showing that compression can relax coverage requirements, while differences in conditional imitation error must also be accounted for.

Assumption 3 can fail even after an arbitrarily small action deviation. For $T \ge 2$, suppose the expert always executes $u_0 = 0$, whereas the learned policy executes an admissible $u_0 = \varepsilon \ne 0$. Histories recording $u_0 = \varepsilon$ then occur under the learned policy but have zero probability under the expert demonstration distribution. No finite coverage constant C can satisfy the assumption.

A representation can merge such distinctions, potentially relaxing coverage requirements, but may also discard information needed to distinguish action consequences. This tension between distributional alignment and information loss under a noninvertible representation also appears in domain-adaptation analyses [17], [18].

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν. In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \le C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event $\mathcal{B}$, with $1 \le C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(h)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(h)^2 \mid Z = z]$. With square-integrable action error under both distributions and measurable versions of these conditional means,

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C_\phi R_\nu(\pi) + D_\phi. \tag{10}$$

If Assumption 4 also holds, the performance-difference identity gives

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + D_\phi}. \tag{11}$$

Appendix A-C provides the proof. With imitation risk and action sensitivity fixed, (11) shows that the effect of a representation on the bound depends jointly on the coverage factor C_ϕ and the conditional error discrepancy D_ϕ. This motivates seeking representations that reduce C_ϕ while controlling D_ϕ, so that their combined contribution $C_\phi R_\nu(\pi) + D_\phi$ decreases.

C. Task-Relevant Information Preservation

We seek conditions under which a representation can improve coverage while controlling the conditional imitation-error discrepancy. We examine semantic guidance as a way to retain task-relevant information, then connect the accuracy of that information to expert-action reconstruction and a bound on D_ϕ.

1) Semantic-Guided Compression and Coverage: Semantic guidance can encourage an encoder to merge appearance differences while retaining object-level structure relevant to action selection. This motivates a representation design, not an automatic coverage guarantee. As in Section III-B3, compression can relax coverage requirements by making some history

distinctions irrelevant to representation events. Appending a deterministic semantic map while retaining the original RGB input does not itself change the best coverage constant relative to RGB, since projection recovers that input. The relevant object is the learned decision representation $Z = \phi(t, \mathcal{H}_t)$, not the number of input modalities.

Improved coverage alone does not ensure a tighter bound; the error associated with merging histories must also be accounted for. A sufficient route is to retain information from which a common rule can reconstruct expert actions under both demonstration and deployment distributions. This emphasis on decision-relevant information is consistent with information-state and behaviorally grounded representation analyses [19], [20]. We next formulate this requirement using object attributes and spatial relationships.

2) Task Information and Expert-Action Reconstruction: We examine when retained task information is sufficient to approximate expert choices. The reconstruction error depends on both what the selected information fails to explain and how accurately it is estimated.

Fix the evaluated policy π and set $\mu = \bar{\mu}_\pi$, with ν the demonstration distribution. Write $\pi(h) = p(Z)$ for the same measurable policy readout under both distributions. We define a task descriptor vector $G \in \mathbb{R}^{d_G}$ whose components describe task-relevant object attributes and spatial relationships. Semantic masks provide a way to construct such quantities, including visibility and relative position. Let χ denote task-specified proprioceptive and execution context, selected in advance rather than identified with the unrestricted history. Suppose Z retains χ and supports an estimate $\widehat{G} \in \mathbb{R}^{d_G}$ of G. For $\rho \in \{\mu, \nu\}$, define $\delta_{G,\rho} = \mathbb{E}_\rho \|G - \widehat{G}\|$. These quantities must be defined under both distributions, with an explicit convention for unavailable components.

Assumption 5 (Descriptor-conditioned expert selection): There exists a common measurable rule g_{E} mapping descriptors and context to $\mathcal{A}$ such that, for $\rho \in \{\mu, \nu\}$,

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - g_{\mathrm{E}}(G, \chi)\| \le \epsilon_{\mathrm{sel},\rho}. \tag{12}$$

For a finite constant $K_{\mathrm{E}} \ge 0$,

$$\|g_{\mathrm{E}}(G, \chi) - g_{\mathrm{E}}(\widehat{G}, \chi)\| \le K_{\mathrm{E}} \|G - \widehat{G}\| \tag{13}$$

holds almost surely under both distributions, with finite $\epsilon_{\mathrm{sel},\rho}$ and $\delta_{G,\rho}$.

Here, $\epsilon_{\mathrm{sel},\rho}$ measures expert choices unexplained by (G, χ), while K_{E} controls sensitivity to estimation error. Task relevance alone does not imply this condition: equally successful actions may reflect different expert choices.

Define $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality yields

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - \widehat{a}_{\mathrm{E}}(Z)\| \le b_\rho := \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}} \delta_{G,\rho}. \tag{14}$$

Appendix B-A gives the derivation. Supervising descriptor readouts targets the estimation term, while the choice of descriptors and context determines how well expert actions can be explained.

3) Bounds on Conditional Imitation-Error Discrepancy: Expert-action reconstruction connects information preservation to the discrepancy in Section III-B3. Because both $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z)$ depend on the same representation, their squared difference is a common function under μ and ν. Expert reconstruction error bounds how far the actual conditional imitation errors depart from this function.

Assume a finite action-set diameter $B_{\mathcal{A}} = \sup_{a,a' \in \mathcal{A}} \|a - a'\|$ under a fixed action scale.

Proposition 1 (Conditional-error transfer through expert reconstruction): Under Assumption 5 and the representation-level coverage condition in Section III-B3, with b_ρ defined in (14),

$$D_\phi \le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu). \tag{15}$$

If Assumption 4 also holds,

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu)}. \tag{16}$$

The proof is in Appendix B-B. The design objective is therefore to improve coverage while preserving information needed to reconstruct expert actions. For a fixed expert baseline, horizon, and action sensitivity, reducing the combined imitation and reconstruction terms tightens the task-failure bound.

IV. Method

Section III motivates preserving information relevant to expert action selection. Figure 1 organizes this approach into two components. First, quality-aware iterative annotation expands a small manually reviewed seed set into semantic supervision. Second, RGB observations, semantic maps, and supervised task-descriptor queries jointly support a base action-chunk policy.

A. Quality-Aware Iterative Semantic Supervision

Task-descriptor labels are derived from semantic masks, so segmentation errors can compromise the supervision used to preserve task information. We combine reviewed seed annotations with video-level quality screening to limit error propagation as the labeled set expands (Fig. 2).

1) Event-Guided Seeds and Iterative Expansion: We select a small set of videos spanning the principal interaction and occlusion states, and annotate keyframes around events such as target exposure, contact, and placement. Bounding boxes and positive and negative point prompts guide bidirectional SAM2 propagation [21]; consistency checks, class-dependent structural processing, and manual correction produce the seed masks. The seeds train an initial U-Net segmenter, which predicts candidate masks on new unlabeled videos. Accepted masks accumulate across iterations, and each subsequent segmenter is reinitialized and trained on this cumulative pool without the original seed videos. The seed videos provide a fixed reference for label agreement, while held-out videos monitor temporal anomalies. Both are tracked across iterations to assess whether expansion compromises annotation quality.

2) Video-Level Quality Screening: The screening procedure distinguishes sustained task changes from transient segmentation anomalies. For each class, we extract trajectories of mask area, spatial extent, centroid, and topology, and fit robust piecewise trends over the video. Persistent changes can establish new segments, while short-lived departures provide anomaly evidence instead of penalizing every inter-frame

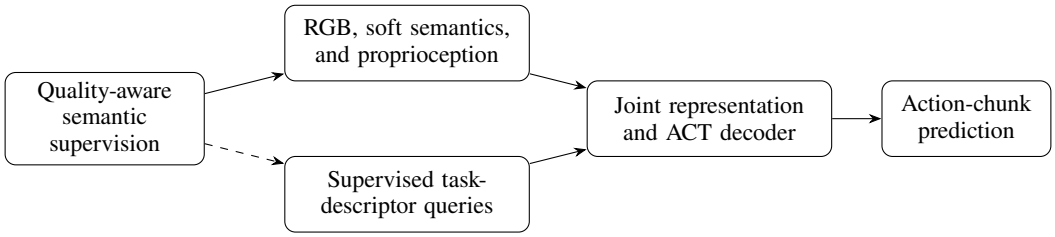


Fig. 1. Method overview. Quality-aware iterative annotation provides a frozen segmenter and quality-weighted descriptor labels. RGB, soft semantics, proprioception, and supervised queries jointly support action-chunk prediction.

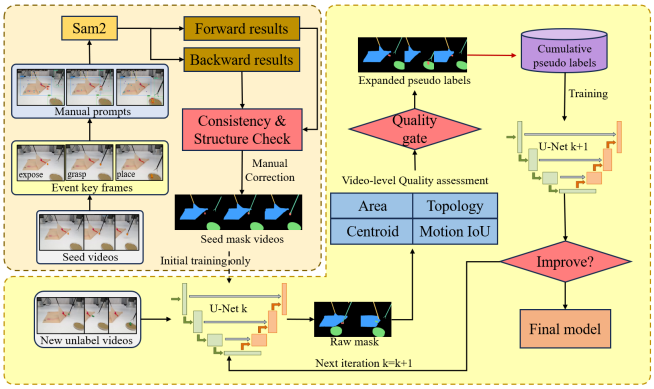


Fig. 2. Quality-aware iterative annotation. Reviewed bidirectional SAM2 masks train the initial segmenter. Subsequent models learn from cumulative, quality-screened pseudo-labels; seed videos are retained for reference rather than further training.

change. Motion-compensated mask IoU supplies additional temporal evidence on pixels with forward–backward-consistent optical-flow correspondences. For class c at frame t, we collect these diagnostics as

$$\boldsymbol{\upsilon}_{t,c}^{\text{seg}} = (\upsilon_{t,c}^{\text{area}}, \upsilon_{t,c}^{\text{space}}, \upsilon_{t,c}^{\text{topo}}, \upsilon_{t,c}^{\text{motion}}). \tag{17}$$

Agreement across evidence groups identifies anomalous intervals; excessive changes introduced by structural processing also trigger exclusion. A frame enters subsequent segmentation training only when its relevant classes pass these checks. The manually reviewed reference set complements temporal screening by checking for persistent semantic errors that temporal consistency alone cannot detect. The same diagnostics also determine descriptor-specific supervision weights in Section IV-B. Quality assessment is performed offline; the final per-view segmenters are frozen and require only current images during policy execution.

B. Semantic Inputs and Task-Descriptor Queries

Semantic maps emphasize object-level spatial structure, while supervised queries encourage the joint representation to retain task-relevant attributes and relationships. RGB and proprioception remain available for decision cues not captured by the semantic classes. This design targets the descriptor-estimation term in (14) through the learned representation, rather than assuming that adding semantic channels improves coverage.

1) Joint Visual-Semantic Encoding: For view v, a frozen segmenter predicts class probabilities $p_t^v(c,\xi)$ from image I_t^v. We form a semantic image from the filtered class labels:

$$\bar{c}_t^v = \mathcal{T}_v\left(\arg\max_{c\in\mathcal{C}_v} p_t^v(c,\cdot)\right), \qquad S_t^v(\xi) = e_{\bar{c}_t^v(\xi)}^{\text{sem}}, \tag{18}$$

Here, $\mathcal{C}_v$ is the view-specific class set, ξ a pixel, $\mathcal{T}_v$ the class-dependent structural filter, and $e_c^{\text{sem}} \in [0,1]^3$ a fixed class color. Offline masks provide these images during policy training; frozen segmenters provide them at deployment. The fixed color representation allows RGB and semantics to share a three-channel visual encoder. The encoded streams from all views are combined with proprioceptive and query tokens in a transformer encoder. An ACT decoder [6] reads this context to predict the base action chunk.

2) Supervised Descriptor Queries: We select K_{rel} descriptor groups $G^{(j)} \in \mathbb{R}^{d_j}$, such as visible area and relative position, which together form the task descriptor vector G. Fixed geometric operators applied to the offline masks generate pseudo-labels $\widetilde{G}_t^{(j)}$; the task-specific definitions are given in Appendix C-C. Each group has a shared learnable query whose encoded output $z_t^{(j)}$ supports view-specific heads, $\widehat{G}_t^{(j,v)} = D_{j,v}(z_t^{(j)})$, and remains available to the action decoder. The view-specific outputs are concatenated rather than averaging geometry across camera coordinates. Supervision therefore encourages task information to be recoverable from the policy representation without prescribing an attention map.

A weight $\varpi_{n,j,v}^G \in [0,1]$ combines descriptor validity with quality evidence from the relevant semantic classes. Reliable labels receive full weight, uncertain labels are downweighted, and undefined or anomalous labels receive zero weight. In particular, missing objects invalidate geometric relations, while a zero-visible-area label remains eligible subject to quality screening. Disabling a descriptor loss does not remove the sample's action supervision. With descriptors normalized to fixed scales, the training objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{K_{\text{rel}} V} \sum_{j,v} \frac{\sum_n \varpi_{n,j,v}^G \ell_G(\widehat{G}_n^{(j,v)} - \widetilde{G}_n^{(j,v)})}{\max(1, \sum_n m_{n,j,v}^G)},$$
$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{IL}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}. \tag{19}$$

Here, n indexes minibatch samples, V is the number of views, $m_{n,j,v}^G$ indicates geometric validity, and ℓ_G is the component-averaged Smooth L1 loss. The denominator counts geometrically valid labels rather than their quality weights, so poor-quality labels reduce the auxiliary supervision strength. $\mathcal{L}_{\text{IL}}$ is

the standard ACT training objective. The segmenters remain frozen, and quality weights affect auxiliary supervision rather than online policy inputs. Appendices C-A and C-B specify label admission and its connection to descriptor-estimation error.

V. Experiments

本节检验语义引导的任务信息表示能否改善有限示范下的实际任务完成表现。双臂暴露与执行要求机器人主动改变遮挡、建立工具接触并完成目标转移，能够同时考察感知变化与多阶段动作衔接。实验依次考察标注扩展过程的质量变化、完整方法的任务表现、语义输入与受监督查询的贡献，以及不同物体、位置和执行阶段上的收益。评价以实际任务结果而非训练误差为依据，对应第 III-B节对模仿目标与任务目标的区分。

A. Experimental Setup

1) 任务与示范: 平台由两台 SO-ARM101 机械臂及固定的前、侧视相机构成，图像分辨率为 640×360。机器人首先拨开遮挡布，使目标暴露并与布料分离，再使用末端工具将目标送入指定区域，最后恢复布料在前视画面中的覆盖面积。布料恢复不是重新遮盖目标物体。示范集包含方块、纸球和螺丝三类物体，共 108 条轨迹、61,740 帧，以 30 Hz 记录双视角图像、10 维关节状态和关节指令。所有方法使用这组动作示范，不增加策略交互训练数据。

一次任务须在 40 s 内使目标完全位于指定区域内，并使布料在前视图中的像素占比恢复至 0.32 以上。统一前视评估器检查暴露、分离、入区与布料恢复条件，以最近 7 次评估中的至少 4 次成功票确认完成；最终结果采用操作者确认后保存的标签。这将第 III-A节的任务完成判据具体化为可由视觉检查的条件，但不要求场景的全部物理状态可观测。

2) 测试条件与评价指标: 测试使用三类示范物体及一个未出现在动作示范中的蜡笔类别。每类物体在 4×3 网格的 12 个初始位置各测试两次，共 48 个物体–位置条件、每方法 96 次试验。横坐标 $x = 0, \ldots, 3$ 从图像左侧向右侧增加，纵坐标 $y = 0, \ldots, 2$ 从上向下增加。五种方法合计 480 次有效试验；当前记录用于评价跨位置表现，不将未经核实的训练位置划分表述为未见位置泛化。

主要指标为任务成功次数及成功率，辅助指标为完成时间、阶段到达次数与同条件重复成功情况。总体成功率报告 Wilson 95% 区间；方法差异按物体–位置分区组检验，并对 10 个总体两两比较作 Holm 校正。原始记录审计、统计条件和详细区组结果见附录 D。

3) 对照与模块配置: 采用 ACT [6] 和 Diffusion Policy（DP） [8] 作为动作生成基线。在同一 ACT 骨干上，以 S 表示额外语义视觉输入，以 Q 表示受监督的任务描述量查询，构造表 I的四格对照。所有配置均保留 RGB 与本体信息；ACT+Q 只在训练时使用由语义标签构造的辅助监督，部署时不输入分割图。ACT+S 与完整方法使用同一套冻结分割器。

四种 ACT 配置均训练 100,000 步，batch size 为 8，学习率为 10^{-5}，随机种子为 1000。部署时每次预测 60 步、执行前 30 步后同步重新规划。DP 使用 100,000 步 checkpoint，预测 64 步、执行 30 步，采用异步推理；DDIM 在三类示范物体上使用 16 步，在蜡笔上使用 24 步。因此，ACT 四格对照用于分析表示模块的作用，DP 比较反映其所用部署配置的系统表现。所有方法目标控制频率均为 30 Hz，但不假定实际推理延迟完全相同。

TABLE I
Factorial ablation of semantic inputs (S) and supervised task-descriptor queries (Q). All variants retain RGB and proprioception.

Variant	Semantic input	Supervised queries
ACT	–	–
ACT+S	Yes	–
ACT+Q	–	Yes
Ours ($S + Q$)	Yes	Yes

完整方法使用过滤后的硬语义彩色图，与 RGB 共享视觉编码器；训练读取离线标注，推理使用冻结 U-Net。三个共享查询分别描述目标可见面积、工具–目标向量和目标–区域距离，并以独立的双视角读出头产生八维监督量。质量权重与查询监督共同启用；本轮未将其拆分为独立实验因素。具体几何、损失及记录名称映射见附录 D-C。

B. Quality-Aware Iterative Annotation

从少量审核种子出发，迭代流程扩展了可用语义监督，并降低了保留视频上原始预测的异常率，但改善并非逐轮单调。每类示范物体选取一条双视角轨迹作为人工种子，共 3 条轨迹，每个视角 1,967 帧；另保留 3 条轨迹、每视角 1,878 帧监测迭代，不参与分割训练。前、侧视角分别训练 U-Net，种子仅用于初始模型 R0；后续三轮累计覆盖 15、44、102 条非种子轨迹，旧轮准入标签与新轮标签共同训练下一模型。至 R3，前、侧视角分别累积 47,766 和 42,010 帧准入伪标签，包含训练与内部验证划分（图 3(a)）。

在固定保留视频上，前视原始预测的异常率从 R0 的 7.72% 降至 R3 的 4.42%，侧视从 12.19% 降至 9.53%。这里的异常表示面积、空间形状、拓扑和运动一致性中的多类证据同时触发，并不等同于人工确认的分割错误。结构过滤后的侧视异常率从 8.20% 降至 6.34%，但前视从 1.81% 变为 1.92%，没有相同的改善（图 3(b, c)）。两个视角的原始异常率均在 R2 低于 R3，说明扩大伪标签池不保证所有质量指标持续改善。

人工种子还用于检查像素标签一致性，逐轮结果见表 III。R1–R3 的前视前景 mIoU 为 97.90%–97.87%，侧视约 97.66%–97.65%；但侧视 R0 在自身训练种子上为 99.06%，至 R3 下降约 1.41 个百分点。种子受初始教师训练影响，而保留视频没有人工像素真值，因此这些结果分别反映种子参考一致性和无标签异常诊断，不能证明独立测试集上的精度不退化。后续策略使用已导出的 R3 分割器，不将其表述为各项指标均最优。

这组记录展示了少量人工种子向大规模伪标签扩展的可行性及质量监测的必要性。它对应附录 C-B中对伪标签误差的关注，但异常率并非描述量标签误差的直接测量；质量筛选的独立贡献仍需无筛选对照。评价划分与门控口径见附录 D-A。

C. Overall Task Performance

完整方法在 96 次试验中成功 92 次，成功率为 95.83%，相比 ACT 和 DP 分别提高 39.58 和 43.75 个百分点（表 II）。总体区间及分类别可视化见附录图 8。按相同物体–位置条件比较，完整方法相对两种基线和两个单

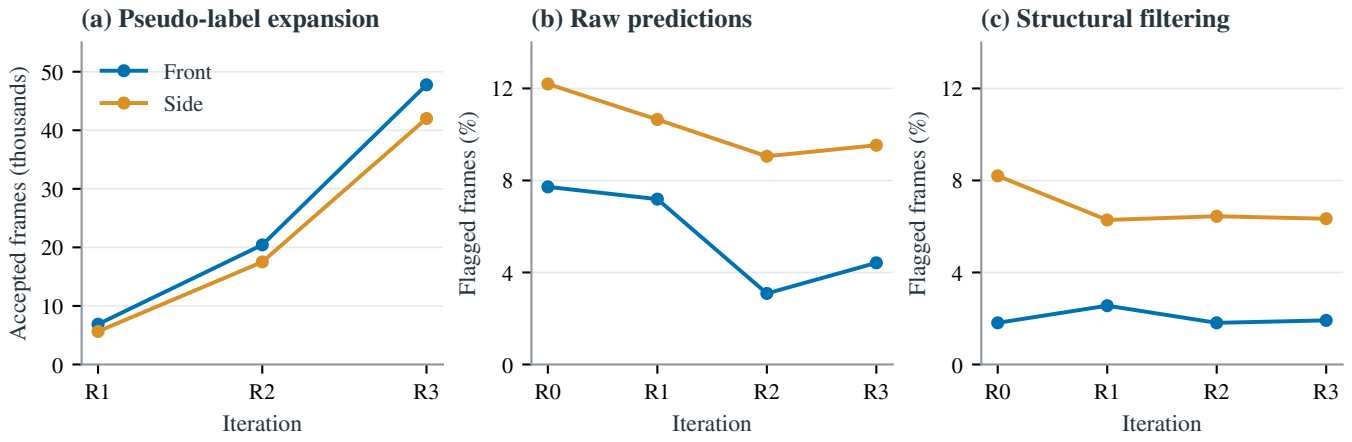


Fig. 3. Annotation expansion and held-out video diagnostics. (a) Cumulative accepted pseudo-label frames, including the internal validation split. (b, c) Automatically flagged frames before and after structural filtering, evaluated on the same three held-out trajectories (1,878 frames per view). R0 is the seed-trained model; R1–R3 use cumulative pseudo-labels. Filtering modifies masks, rather than removing frames from these evaluation denominators.

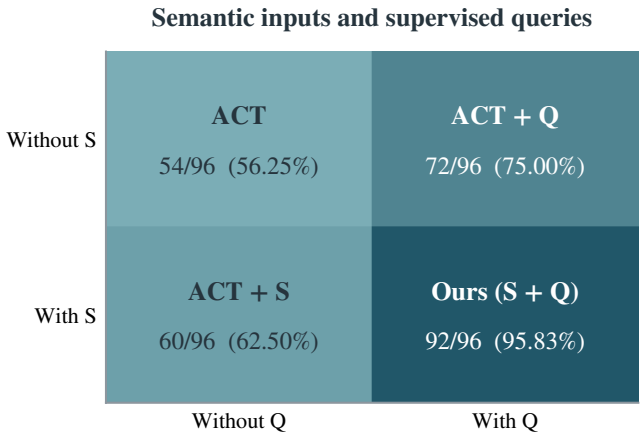


Fig. 4. The 2×2 ACT ablation. Each cell contains 96 trials. The query factor includes task-query tokens and their quality-weighted descriptor supervision.

模块配置的总体差异均在 Holm 校正后达到 $p < 0.001$。详细效应量与检验值见表 IV。

这一比较表明，在相同动作示范和测试条件下，联合表示设计能够改善完整任务结果，而不仅是改变监督损失。不过，总体优势不能单独归因于语义输入或查询监督；下面通过四格消融分析两者的作用。

D. Contributions of Semantics and Supervised Queries

联合配置的收益大于任何一个单模块配置，但仅增加语义输入的改善有限（图 4）。ACT+S 成功 60/96 次，相比 ACT 增加 6 次（6.25 个百分点），总体差异未达到校正后 0.05 水平（$p = 0.817$）。相比之下，ACT+Q 成功 72/96 次，增加 18 次（18.75 个百分点，$p = 0.0188$）。这说明，受监督的任务查询模块在没有额外语义输入时也能带来收益；增加一个语义视觉流并不自动产生相同效果。

当已有查询监督时，增加语义输入使成功次数从 72 提高到 92；当已有语义输入时，增加查询模块使成功次数从 60 提高到 92。两个单模块配置均未达到联合配置的表现。两条增益之差为 14.58 个百分点，作为描述性结果提示联合使用的价值，但本轮没有对该交互量进行独立显著性检验。

这一现象与第 III-C节的设计动机一致：有用的表示不仅需要突出对象结构，还应保留动作选择所需的任务信息。语义图提供对象级输入，而查询监督约束其联合表示中可读出的可见性和空间关系。四格结果支持这套联合设计，但不直接测量描述量误差 $\delta_{G,\rho}$，也不将增加查询的效果等同于已经证明 D_ϕ 下降。

E. Object and Position Variation

完整方法的高成功率并非仅来自某一种物体或少数初始位置。四类物体均成功 23/24 次；三类示范物体合计 69/72 次，示范未见的蜡笔为 23/24 次。在蜡笔上，ACT、DP、ACT+S 及 ACT+Q 分别成功 13、9、17 和 19 次。因此，当前证据支持向这个额外物体类别的迁移，但其范围不是任意未见物体或跨任务泛化。

图 5显示，完整方法在 12 个位置均至少成功 7/8 次，而 ACT 和 DP 的最差位置分别为 1/8 和 0/8。ACT+S 与 ACT+Q 仍分别存在 1/8 和 3/8 的位置。完整方法相对 ACT+Q 新增的 20 次成功中，17 次来自左侧两列；相对 ACT+S 新增的 32 次成功中，30 次来自上方两行。这些空间分布表明，联合配置的改善主要补足了单模块方案表现较弱的区域，而非只在原本容易的位置重复获益。

以每个物体–位置的两次重复为单位，完整方法在 48 个条件中的 44 个条件两次均成功，其余 4 个条件各成功一次，没有两次均失败的条件。ACT、DP、ACT+S 和 ACT+Q 两次均成功的条件数分别为 17、18、22 和 31。每条件仅两次重复不足以精确估计局部成功概率，但与空间分布一起支持完整方法在本测试网格上的一致性。完整物体–位置矩阵见附录图 9。这里检验的是部署行为在条件变化下的表现，不以网格成功率代替第 III-B3节的表示覆盖常数。

F. Execution Bottlenecks and Completion Time

阶段记录将主要收益定位在目标分离后的入区过程，而非仅在初始暴露阶段。ACT 在 89 次试验中曾使物体分离，但仅 55 次曾入区；ACT+S 相应为 93 次和 62 次，ACT+Q 为 92 次和 73 次。完整方法则在全部 96 次试验

TABLE II
Real-robot task performance. Entries report successes/trials (percentage); crayon is absent from the policy demonstrations. Time is averaged over successful trials only.

Method	Cube	Paper ball	Screw	Crayon (unseen)	Overall	Time (s)
ACT	18/24 (75.0%)	12/24 (50.0%)	11/24 (45.8%)	13/24 (54.2%)	54/96 (56.2%)	14.96
DP	12/24 (50.0%)	16/24 (66.7%)	13/24 (54.2%)	9/24 (37.5%)	50/96 (52.1%)	15.28
ACT + Q	18/24 (75.0%)	16/24 (66.7%)	19/24 (79.2%)	19/24 (79.2%)	72/96 (75.0%)	13.55
ACT + S	16/24 (66.7%)	11/24 (45.8%)	16/24 (66.7%)	17/24 (70.8%)	60/96 (62.5%)	13.80
Ours (S + Q)	**23/24 (95.8%)**	**23/24 (95.8%)**	**23/24 (95.8%)**	**23/24 (95.8%)**	**92/96 (95.8%)**	**14.77**

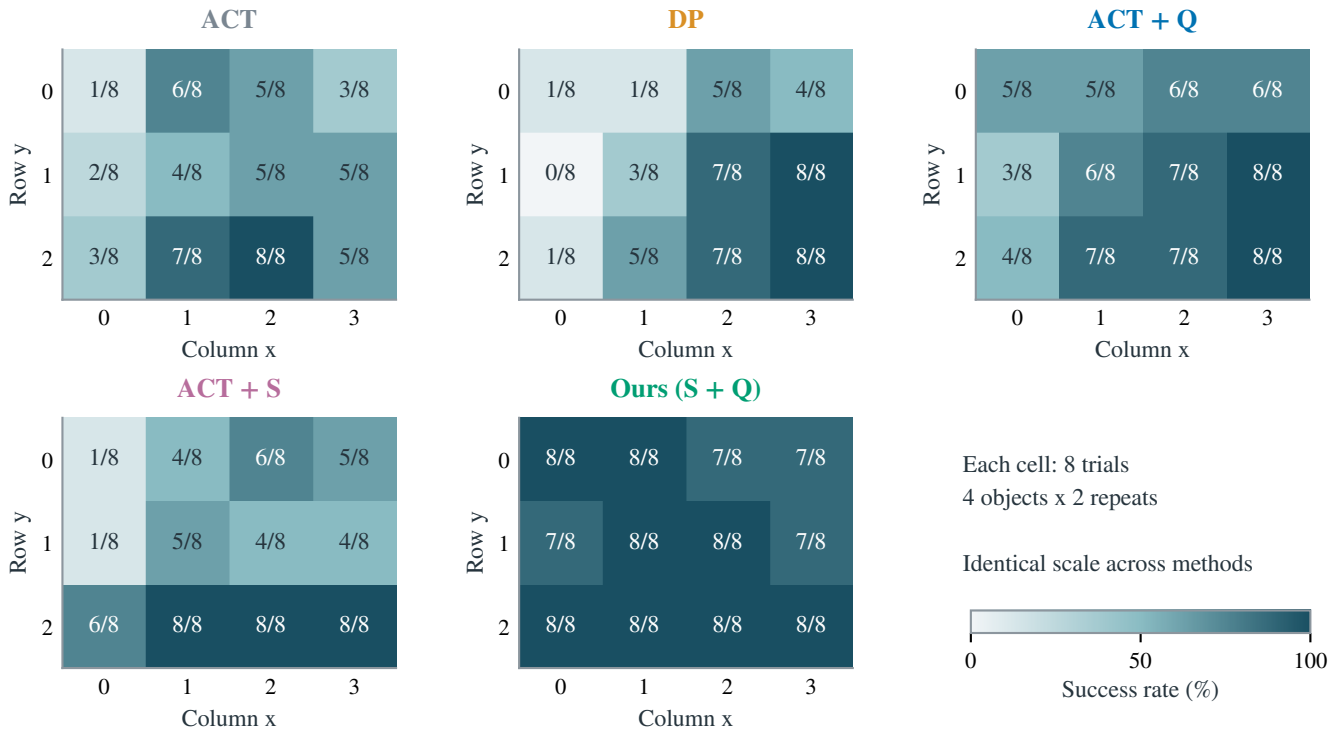


Fig. 5. Success across the initial-position grid. Each cell pools four objects and two repeats ($n = 8$); all panels share the same 0–100% scale. Coordinates refer to the image grid, not physical depth.

中达到分离，92 次入区并最终成功。因此，相比只输入语义图，完整方法最明显的变化是减少了分离后未能完成目标转移的试验。

ACT+S 的 36 次失败中，31 次发生在已分离但未入区的情形；完整方法的 4 次失败也集中于此（图 6及表 VI）。工具–目标向量和目标–区域距离正是为这一过程选择的描述量。失败分布与这种设计意图相符，但由于三个查询一同启用，不能将收益分别归因于某个查询，或由阶段日志判断具体接触失效原因。

完整方法在 20 s 内成功 86/96 次，30 s 内完成全部 92 次成功试验；ACT 和 DP 在 20 s 内分别成功 48 次和 40 次。这说明收益在统一时间预算内体现为更多完整任务，而不只是等待更久。成功试验平均耗时为 14.77 s，与 ACT 的 14.96 s 和 DP 的 15.28 s 接近；ACT+Q 和 ACT+S 分别为 13.55 s 和 13.80 s，但各方法成功子集不同，不能据此建立公平的速度排名。所有试验均计入时，完整方法的平均控制时长为 15.82 s，主要反映超时失败更少。

这一收益伴随额外感知开销。完整方法的可训练参数比 ACT 增加 7,176 个（约 0.014%），冻结分割器使总参数量增至 76.53 M；日志中的动作块推理流水线均时为 79.24 ms，ACT 为 12.16 ms。这些值包含各自流水线开销，并非查询模块的独立耗时；完整部署配置和时间统计见附录 D-D。

G. Connection to the Analysis

本轮实验提供了三个层面的证据：迭代记录展示语义监督的扩展及其质量变化，整体比较验证任务层面的收益，语义–查询四格对照与阶段分布支持任务信息表示的设计动机。它们与第 III-C3节从任务描述量到专家动作重构、再到条件误差迁移的分析相衔接，但不构成对其假设或上界数值的直接验证。当前没有独立测量 C_ϕ、D_ϕ、部署描述量误差或专家选择近似误差，也没有拆分质量筛选与查询监督的贡献。因此，本文据此报告联合设计在所测任务上的有效性，而不声称已由实机成功率证明理论保证或标注流程的独立因果作用。

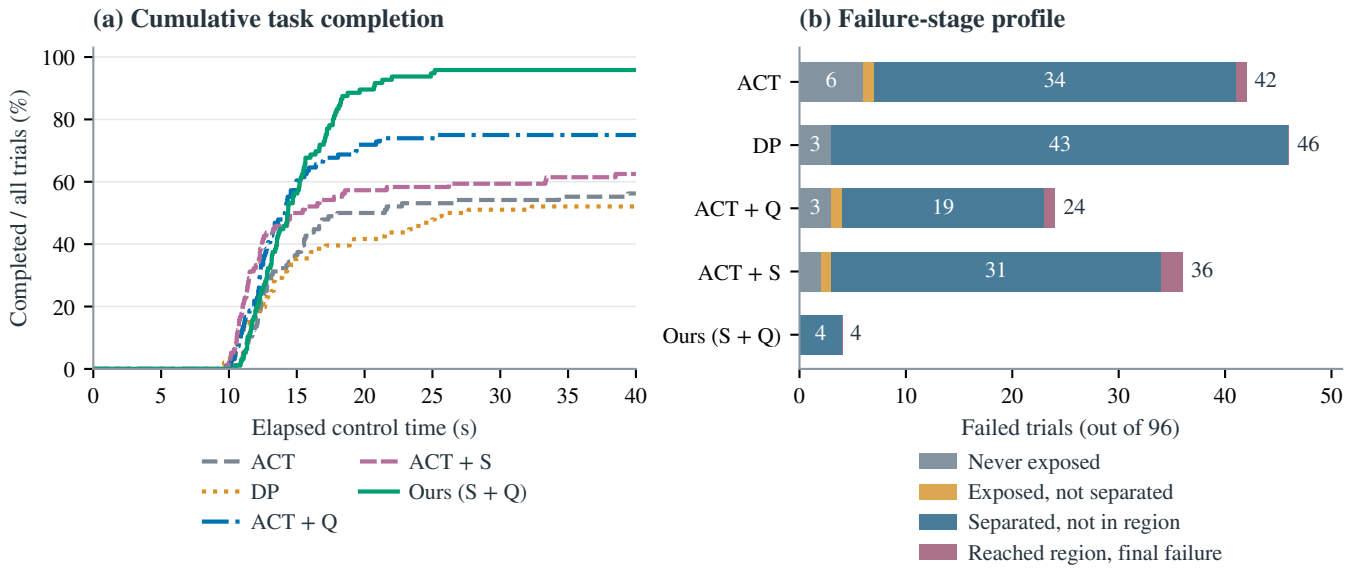


Fig. 6. Completion and failure profiles. (a) Cumulative successful trials divided by all 96 trials; failures remain in the denominator. (b) Mutually exclusive failure categories based on the furthest automatically detected stage and the final reviewed label.

VI. Discussion and Conclusion

A. Failure Cases and Limitations

本节结合表 VI的失败分布和同步视频、指令、感知日志，分析完整方法尚未解决的问题。案例选择覆盖不同失败类型，不以个别成功片段替代整体统计。

1) Perception and Task Information: 质量筛选针对视频中的异常变化，不能排除时间上稳定的系统性误分。所选描述量刻画对象属性与空间关系，但不一定足以解释接触状态或专家的具体动作选择；语义分割准确与描述量读出准确也不意味着策略已经正确利用这些信息。

（结合图 7(a,b) 区分分割错误、描述量估计错误和执行决策错误。说明是否存在描述量准确仍失败的情况，并联系理论中的专家选择近似误差解释未被当前信息覆盖的动作选择因素。没有探针或干预证据时，将原因表述为待验证解释，而非确定归因。）

2) Scope of the Evidence: 本文的任务表现界使用总体模仿误差，并依赖覆盖、动作敏感性和专家选择条件。独立轨迹上的读出和动作误差用于检验相应机制，不能将有限训练损失直接代入这些界，也不能据更紧的上界确定实际成功率排序。两个任务上的分别训练验证方法的适用性，不等同于零样本跨任务迁移；手术模拟材料上的实验亦不构成临床有效性证据。

（根据真实划分与训练重复说明结论覆盖哪些物体、位置和平台。联系理论 B–C 总结哪些机制获得实验支持、哪些条件尚未得到直接测量；概括新增任务的描述量设计、语义标注与训练成本。若未做多预算实验，不声明样本效率优势；QToken 没有独立停止头时，不将任务成功等同于自主停止能力。）

B. Conclusion

（用一段总结问题、两个方法模块及两个实机任务支持的核心结论。填写实际成功率、稳定性或恢复收益，只保留与匹配对照相符的表述；最后指出一个由失败分析直接支持的主要局限，不预先声明显著提升、通用性或收敛保证。）

(a) 持续分割错误：RGB、语义预测、人工审核
(b) 描述量准确但执行失败：任务关系与接触状态
填写两个任务的代表性案例；只标注日志与视频支持的事实

Fig. 7. Representative failure cases and supporting evidence. Placeholder for perception and task-information limitations.

Appendix A

A. Proof of the Policy Performance-Difference Identity

We prove (8) by summing expected changes in the reference value. The evaluation-domain restriction in Section III-A and Assumption 1 give the terminal value

$$V_T^{\pi_0}(\mathcal{H}_T) = 1 - \mathcal{V}(I_{0:T}^{1:V}) = 1 - \Psi. \tag{20}$$

For $t < T$, taking action $\pi(\mathcal{H}_t)$ and then following π_0 gives the failure probability $V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})$ conditional on the next history. Averaging over next histories under Assumption 2 gives

$$Q_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = \mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]. \tag{21}$$

Here, π determines the next-history distribution, while $V_{t+1}^{\pi_0}$ accounts for subsequent execution under π_0.

Combining (7) and (21) and taking expectations gives

$$\begin{aligned}&\mathbb{E}_\pi[A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))]\\ &= \mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)].\end{aligned} \tag{22a}$$

By the tower property, averaging the conditional next-step value over current histories gives its unconditional expectation:

$$\begin{aligned}&\mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right]\\ &= \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})].\end{aligned} \tag{22b}$$

Substituting (22b) into (22a) and summing over time leaves only the terminal and initial terms:

$$\begin{aligned}&\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))\\ &= \sum_{t=0}^{T-1}\left\{\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)]\right\}\\ &= \mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] - \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)].\end{aligned} \tag{22c}$$

At time T, no action remains. By (20), the terminal term is the failure probability under π:

$$\mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] = \mathbb{E}_\pi[1 - \Psi] = F(\pi). \tag{22d}$$

The policies share the initial-history distribution, so the initial term equals the failure probability of following π_0 throughout:

$$\begin{aligned}\mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)] &= \mathbb{E}_{\pi_0}[V_0^{\pi_0}(\mathcal{H}_0)]\\ &= F(\pi_0).\end{aligned} \tag{22e}$$

Finally, substituting (22d) and (22e) into (22c) gives

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = F(\pi) - F(\pi_0). \tag{22f}$$

This establishes (8).

B. Proof of the Task-Failure Bound

We prove (9) by first bounding task consequences through action deviations and then transferring the resulting deployment error to the demonstration distribution.

Set $\pi_0 = \pi_{\mathrm{E}}$ and let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$. Applying Assumption 4 in (8) bounds each step's increase in failure probability by its action deviation:

$$F(\pi) - F(\pi_{\mathrm{E}}) \le L\,\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right]. \tag{23a}$$

Since $\bar{\mu}_\pi$ samples control steps uniformly, the time sum equals T times the deployment average:

$$\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right] = T\,\mathbb{E}_{(t,h)\sim\bar{\mu}_\pi} e(h). \tag{23b}$$

By the Cauchy–Schwarz inequality, the mean action deviation is bounded by its root mean square:

$$\mathbb{E}_{\bar{\mu}_\pi} e(h) \le \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(h)^2}. \tag{23c}$$

Assumption 3 also bounds expectations of nonnegative measurable functions, by simple-function approximation and monotone convergence. Applying it to $e(h)^2$ and using (3) gives

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C\,\mathbb{E}_\nu e(h)^2 = C R_\nu(\pi). \tag{23d}$$

Combining (23a)–(23d) establishes (9).

C. Proof of the Representation-Level Transfer Bound

We prove (10) by separating conditional error discrepancy from distributional coverage, then apply action sensitivity to obtain (11).

Let $\mu = \bar{\mu}_\pi$, with the remaining notation and the integrability and measurable conditional-mean conditions as in Section III-B3. Representation coverage ensures that ν_Z-null sets are also μ_Z-null sets, so the demonstration-side conditional mean is well defined for the deployment comparison.

First, conditioning on the representation gives

$$\mathbb{E}_\mu e(h)^2 = \mathbb{E}_{\mu_Z} m_\mu(Z). \tag{24a}$$

At the same representation, deployment and demonstration conditional errors can differ. By the definition of D_ϕ,

$$\begin{aligned}\mathbb{E}_{\mu_Z} m_\mu(Z) &\le \mathbb{E}_{\mu_Z} m_\nu(Z) + \mathbb{E}_{\mu_Z}|m_\mu(Z) - m_\nu(Z)|\\ &= \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi.\end{aligned} \tag{24b}$$

Finally, applying representation coverage to the nonnegative function m_ν and taking the total expectation give

$$\mathbb{E}_{\mu_Z} m_\nu(Z) \le C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) = C_\phi R_\nu(\pi). \tag{24c}$$

Combining (24a)–(24c) establishes (10). Under Assumption 4, substituting this result into the sensitivity, time-average, and Cauchy–Schwarz steps (23a)–(23c) yields (11).

Appendix B
Proofs for Task-Relevant Information Preservation

A. Proof of the Expert-Action Reconstruction Bound

We prove (14) under the common encoder and readouts of Section III-C, with $\mu = \bar{\mu}_{\pi}$. Define the common represented expert rule $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality and Assumption 5 give, for $\rho \in \{\mu, \nu\}$,

$$\begin{aligned}\mathbb{E}_{\rho}\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| &\le \mathbb{E}_{\rho}\|\pi_{\mathrm{E}} - g_{\mathrm{E}}(G,\chi)\| \\ &\quad + \mathbb{E}_{\rho}\|g_{\mathrm{E}}(G,\chi) - g_{\mathrm{E}}(\widehat{G},\chi)\| \\ &\le \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}}\delta_{G,\rho} = b_{\rho}.\end{aligned} \tag{25}$$

B. Proof of the Conditional-Error Transfer Bound

We prove Proposition 1 by comparing actual imitation error with a common function of Z, then transferring the demonstration-side reconstruction error through representation coverage. Use $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$ from Section III-C. Since $\pi(h) = p(\phi(t,h))$, the represented squared error $\ell_Z(z) = \|p(z) - \widehat{a}_{\mathrm{E}}(z)\|^2$ is the same function of z under both distributions. All three actions belong to $\mathcal{A}$, so the reverse triangle inequality yields

$$\begin{aligned}|e(h)^2 - \ell_Z(Z)| &\le \big(\|\pi - \pi_{\mathrm{E}}\| + \|\pi - \widehat{a}_{\mathrm{E}}\|\big)\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \\ &\le 2B_{\mathcal{A}}\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\|.\end{aligned} \tag{26}$$

Conditioning separately under each distribution gives

$$|m_{\rho}(z) - \ell_Z(z)| \le 2B_{\mathcal{A}}\mathbb{E}_{\rho}[\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \mid Z = z]. \tag{27}$$

Insert the common function ℓ_Z between m_{μ} and m_{ν} and apply representation coverage to the nonnegative demonstration-side conditional error:

$$\begin{aligned}D_{\phi} &\le \mathbb{E}_{\mu_Z}|m_{\mu}(Z) - \ell_Z(Z)| \\ &\quad + \mathbb{E}_{\mu_Z}|m_{\nu}(Z) - \ell_Z(Z)| \\ &\le \mathbb{E}_{\mu_Z}|m_{\mu}(Z) - \ell_Z(Z)| \\ &\quad + C_{\phi}\mathbb{E}_{\nu_Z}|m_{\nu}(Z) - \ell_Z(Z)| \\ &\le 2B_{\mathcal{A}}(b_{\mu} + C_{\phi}b_{\nu}).\end{aligned} \tag{28}$$

Representation coverage ensures that the demonstration-side conditional inequality also holds μ_Z-almost everywhere. This proves (15); substitution into (11) establishes (16).

Appendix C
Method Details

A. Descriptor Labels and Quality Weights

Descriptor supervision uses the quality evidence relevant to its geometric definition rather than a single frame-wide score. Visible area depends primarily on target extent and topology; relative position also depends on the localization of both objects, and a tool-endpoint vector is sensitive to tool shape. For descriptor j in view v, let $\mathcal{C}_j$ contain its required classes, $\eta^{\mathrm{seg}}_{t,c,v}$ indicate that class c passes anomaly screening, and $m^{G}_{t,j,v}$ indicate geometric validity. The supervision weight is

$$\varpi^{G}_{t,j,v} = m^{G}_{t,j,v} \prod_{c \in \mathcal{C}_j} \eta^{\mathrm{seg}}_{t,c,v} \min_{c \in \mathcal{C}_j} q^{\mathrm{seg}}_{t,c,v}, \tag{29}$$

where $q^{\mathrm{seg}}_{t,c,v} \in [0,1]$ is the exported class-specific quality score. These diagnostic weights are not calibrated probabilities of correctness. Area uses object quality; tool–object vectors use object and tool quality; region distance uses object and region quality. Normalization in (19) counts geometric validity independently of quality, preserving the attenuation of uniformly uncertain labels.

An empty mask supplies a zero-area candidate subject to quality screening; an undetected missed segmentation can still produce an erroneous absence label. Undefined distances and directions are excluded from the auxiliary loss rather than labeled as zero. Missing optical-flow correspondences provide no motion evidence, whereas excessive structural repair excludes a frame from segmentation training. Whole-video diagnostics may use later frames offline; they are not supplied to the deployed policy.

B. Pseudo-Label and Descriptor-Readout Errors

For either evaluation distribution ρ used in Section III-C, inserting the descriptor pseudo-label $\widetilde{G}$ gives

$$\delta_{G,\rho} \le \mathbb{E}_{\rho}\|\widehat{G} - \widetilde{G}\| + \mathbb{E}_{\rho}\|\widetilde{G} - G\|. \tag{30}$$

The terms separate descriptor fitting from pseudo-label error, motivating quality screening in addition to query supervision. When measured only on admitted labels, this comparison concerns the admitted subset; the admission rate is reported alongside the error.

C. Task-Descriptor Instance

Our instance uses three task-shared queries with separate front- and side-view heads. They describe target visibility, tool–target displacement, and target–region distance, producing two, four, and two scalars, respectively. For each view, let $M^{v}_{O}, M^{v}_{R}, M^{v}_{T}$ be the object, region, and tool masks in an image of width W and height H_I. The labels are

$$\begin{aligned}\widetilde{G}^{(1,v)} &= |M^{v}_{O}|/(WH_I), \\ \widetilde{G}^{(2,v)} &= \mathrm{clip}(\bar{c}^{v}_{O} - \bar{r}^{v}_{T}, -1, 1), \\ \widetilde{G}^{(3,v)} &= \frac{\max_{\xi \in M^{v}_{O}} \mathrm{dist}(\xi, \mathcal{P}_v)}{\sqrt{(W-1)^2 + (H_I - 1)^2}}.\end{aligned} \tag{31}$$

Here, $\bar{c} = (c_x/(W-1), c_y/(H_I - 1))$ normalizes a centroid, with rightward and downward positive directions. For the front view, $\bar{r}^{v}_{T}$ is the left endpoint of the tool mask's principal-axis segment in normalized coordinates; for the side view, it is the tool centroid. The fixed polygon $\mathcal{P}_v$ is estimated offline from training masks. Distance is zero inside this polygon, so the third descriptor vanishes when every visible object pixel is inside. Its label requires nonempty object and region masks; the vector requires a nonempty object mask and a valid tool reference. Image-plane containment of visible pixels does not establish containment of an occluded physical object.

TABLE III
Annotation diagnostics (%). Seed mIoU measures raw foreground-mask agreement on teacher-dependent manual references. Raw/filtered columns report flagged-frame rates on held-out videos without pixel ground truth. "–" indicates an unavailable record.

View	Model	Seed mIoU	Raw	Filtered
Front	R0	–	7.72	1.81
Front	R1	97.90	7.19	2.56
Front	R2	97.88	3.09	1.81
Front	R3	97.87	4.42	1.92
Side	R0	99.06	12.19	8.20
Side	R1	97.66	10.65	6.28
Side	R2	97.67	9.05	6.44
Side	R3	97.65	9.53	6.34

D. Evaluation Scope

Seed videos used by the initial segmenter provide a development reference, not an independent final test. Cross-video segmentation evaluation excludes annotation-training videos; an independent final test also excludes model selection. Descriptor-generalization evaluation additionally excludes policy-training videos and uses the base policy in deployment mode. For ACT, this sets the latent input to zero rather than conditioning on the demonstrated action posterior [6].

Appendix D
Evaluation Details

A. Annotation Evaluation Protocol

人工种子轨迹编号为 0、40 和 80，固定保留轨迹为 8、43 和 94，两组均覆盖三类示范物体。其余 102 条轨迹分为 15、29、58 条新视频批次，跨轮累积；两个相机使用相同轨迹划分，不能视为六条独立保留轨迹。种子不直接参与 R1–R3 拟合，但仍通过教师伪标签影响训练；保留轨迹与各模型及其教师的训练来源均不重叠。该保留集用于多轮监测，不是从未查看的最终测试集。

每视角包含背景及六个前景类别：遮挡布、物体、区域、工具、左机械臂和右机械臂。种子 mIoU 先累加三条视频的像素混淆矩阵，再计算六类前景 IoU 的均值，不平均各视频的 mIoU。保留视频异常率为至少一个类别被标记的帧数除以全部 1,878 帧；原始与结构过滤分支分别计算，均不删除异常帧后再评价。各轮使用相同质量算法配置及各视角固定的结构过滤规则；鲁棒阈值仍按各视频、各类别的预测轨迹估计。因此异常率属于内部一致性诊断，稳定但语义错误的掩码可能未被检出。

训练准入另行执行：原始预测中同类至少两组异常证据触发时剔除该帧及前后各两帧；结构过滤改变像素相对原始前景超过 20%，或过滤后前景不足 20 像素时也剔除。本轮未另外启用正的连续质量分数阈值。过滤后的准入掩码跨轮保留，最终每视角候选池为 57,895 帧；前、侧视角准入比例分别为 82.50% 和 72.56%。这些比例描述不同相机的训练数据筛选，不是保留集上的预测精度。

每轮从 ImageNet 初始化重新训练，准入帧内部按 80:20 随机划分训练与验证。相邻帧可能分属两组，因此不以内部验证 mIoU 报告跨视频泛化。本次记录覆盖三轮累计扩展，不构成最优迭代次数或收敛性实验；也未记录人工工时，不能由种子视频比例推算人力节省率。

TABLE IV
Selected overall contrasts. Positive differences favor the first method; p values are Holm-adjusted over all ten pairwise comparisons.

Comparison	Difference (pp)	Adjusted p
ACT + S vs. ACT	+6.25	0.8172
ACT + Q vs. ACT	+18.75	0.0188
Ours (S + Q) vs. ACT	+39.58	3.83×10^{-10}
Ours (S + Q) vs. DP	+43.75	8.57×10^{-12}
Ours (S + Q) vs. ACT + Q	+20.83	1.15×10^{-4}
Ours (S + Q) vs. ACT + S	+33.33	2.21×10^{-8}

B. Trial Audit and Statistical Comparisons

最终数据由五种方法各 96 次有效记录构成，覆盖四类物体、12 个位置和每条件两次重复。ACT+S 在方块 $(1, 0)$ 的一次测试中，自动评估器判为成功，操作者确认目标停在区域边缘并保存为失败；本文按保存标签统计。该组在纸球 $(3, 0)$ 的一次 18.7 s 手动中止已排除，并由正常执行至 40 s 的失败记录替换。补采后 480 条记录的物体–位置–重复编号组合没有缺失或重复；其他四组记录保持不变。

跨方法比较在 48 个物体–位置区组内进行。每个区组的两方法共四个二值结果，在固定总成功数下按每方法两次重新分配；将各区组的超几何分布卷积，得到成功次数差的精确零假设分布。总体两两比较使用统一的 10 项 Holm 校正族，表 IV列出正文使用的对照。该检验依赖区组内结果可交换；试验顺序未随机化，不能排除顺序或环境漂移的影响。Wilson 区间为汇总二值计数的描述性区间，不刻画跨训练种子的波动。每种配置只有一个训练种子；每条件两次执行不是两次独立训练。

C. Tested Policy Configuration

记录名 ACT、DP、RGBQtoken、segNoQtoken 和 viewQtoken 分别对应 ACT、DP、ACT+Q、ACT+S 和 Ours $(S + Q)$。S 增加双视角硬语义彩色图，不替代 RGB；Q 增加三个查询及质量加权描述量监督。三个查询分别输出双视角可见面积（2 维）、工具–目标向量（4 维）和目标–区域距离（2 维）。同一查询共享任务含义，但不同视角由独立读出头监督，不平均不同视角的像素几何量。几何定义见附录 C-C。

训练语义输入来自离线掩码，部署语义输入来自冻结 U-Net 的 argmax 及结构过滤结果；RGB 与语义共享视觉骨干。查询的标量 Smooth L1 损失使用 $\beta = 0.01$，先在每个视角读出头内按几何有效分量数归一化，再平均六个读出头，辅助损失系数为 1。相关类别的质量分数取最小值；任一相关类别被标记为异常，或几何量无定义时，该项辅助监督权重为零。分母不使用质量权重之和，因此整体较差的标签仍会降低辅助监督强度。动作监督不因辅助标签失效而删除。

四种 ACT 配置的动作示范、训练步数、batch size、学习率和种子相同；语义及查询配置额外使用离线标注所提供的信息，不称为相同监督信息量的比较。本轮没有无监督查询、无质量权重或单查询对照，不能将 Q 的收益进一步拆分。DP 的扩散目标及异步部署不同于 ACT；蜡笔测试还增加了 DDIM 步数，应按所报告配置解释其结果。

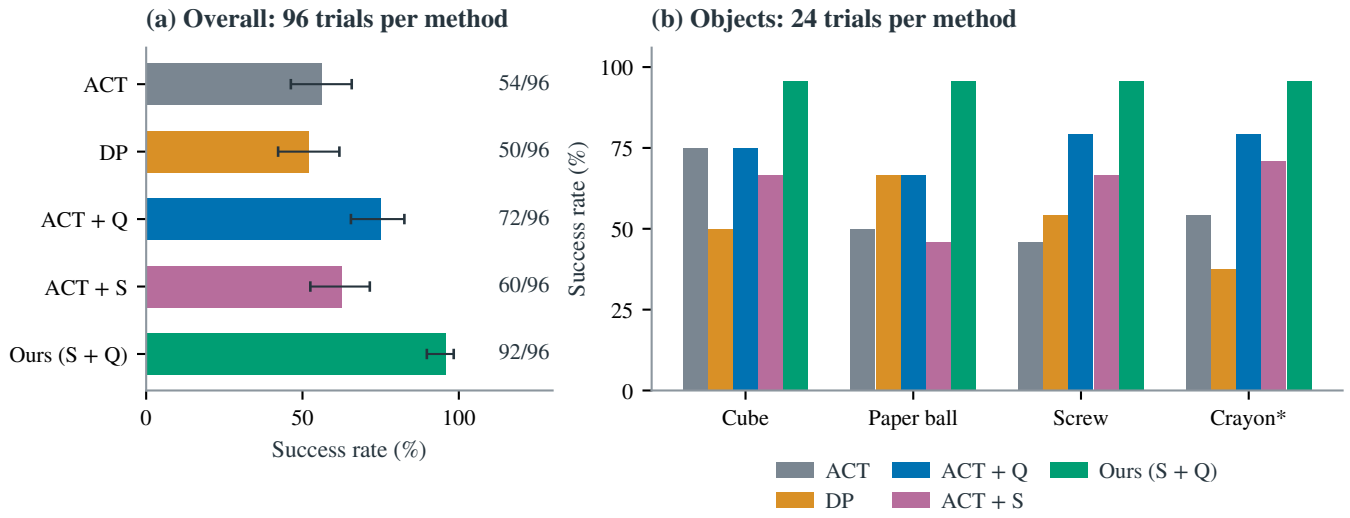


Fig. 8. Overall and object-specific performance. (a) Overall success rates with Wilson 95% intervals. (b) Success rates for each object; crayon (*) is absent from the policy demonstrations.

TABLE V
Overall uncertainty and timing. CI: Wilson 95% interval for success rate; successful/all times are means in seconds.

Method	CI (%)	Successful	All	By 20 s
ACT	[46.3, 65.7]	14.96	25.92	48/96
DP	[42.2, 61.8]	15.28	27.13	40/96
ACT + Q	[65.5, 82.6]	13.55	20.17	69/96
ACT + S	[52.5, 71.5]	13.80	23.43	55/96
Ours (S + Q)	[89.8, 98.4]	14.77	15.82	86/96

TABLE VI
Mutually exclusive failure categories. E: never exposed; S: exposed but not separated; R: separated but not in region; F: reached region but final failure.

Method	E	S	R	F	Total
ACT	6	1	34	1	42
DP	3	0	43	0	46
ACT + Q	3	1	19	1	24
ACT + S	2	1	31	2	36
Ours (S + Q)	0	0	4	0	4

D. Completion Time and Runtime

表 V同时报告成功子集耗时、全部试验耗时及固定时间内完成数。失败试验通常达到 40 s 截止时间；因此全部试验耗时同时反映速度与可靠性，而成功子集均值存在不同的样本选择。累积完成曲线始终以每方法 96 次为分母。

日志中的流水线平均推理时长按记录的推理次数加权：ACT 为 12.16 ms，DP 为 255.48 ms，ACT+Q 为 54.15 ms，ACT+S 为 71.20 ms，完整方法为 79.24 ms。这不是隔离硬件负载后的组件基准，不能将方法间差值全部归因于某个新增模块。ACT 的可训练参数为 51,584,906，完整方法为 51,592,082；完整方法计入冻结分割器后共 76,531,328 个参数。ACT 每次预测 60 步、执行 30 步后同步重新规划，DP 预测 64 步、执行 30 步并异步更新。目标控制频率均为 30 Hz，但流水线推理时间与指令发送频率不是同一指标。

E. Stage and Condition Results

图 8补充总体区间与物体分层结果；图 9展示所有 240 个方法–物体–位置单元，每格两次执行。这些结果描述当前物体和网格，不将小样本单元解释为精确的局部成功概率。

表 VI按最终失败标签和自动检测到的最远阶段划分失败，类别互斥且总和等于失败数。E 表示从未暴露，S 表示已暴露但从未分离，R 表示已分离但从未入区，F 表示曾入区但最终失败。最后一类不等同于单独的布料恢复失败，也不直接给出接触、遮挡或控制层面的根因。阶段到达指试验中曾满足相应条件，不是重新初始化后的独立子任务成功率。

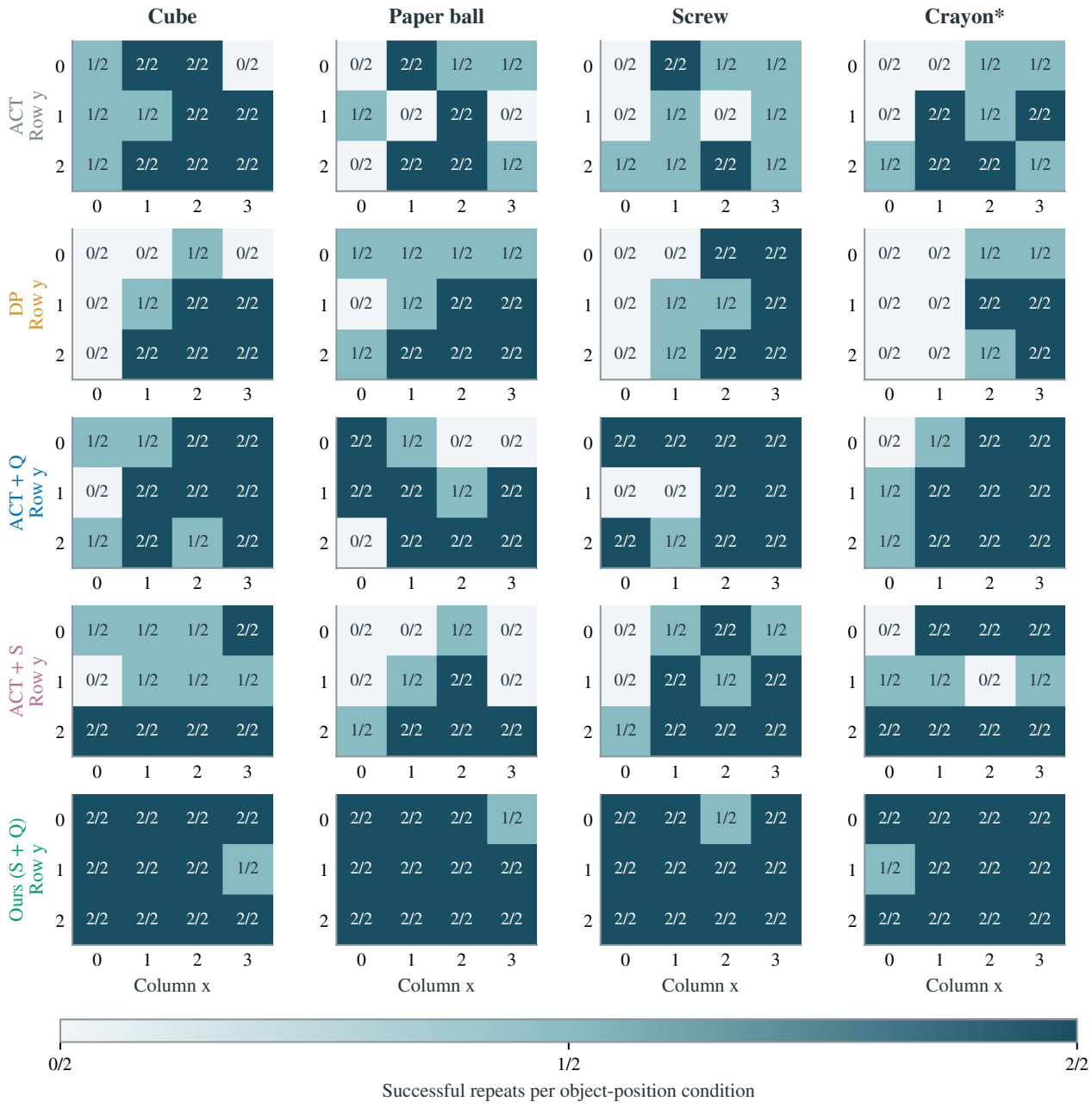


Fig. 9. Complete object–position results. Each panel follows the same four-column, three-row test grid. Cells show successful repeats out of two, with a common scale across methods and objects. Crayon (*) is absent from the policy demonstrations.

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